ModelIt!

Teacher's Guide

The Carbon Bomb: Why Thawing Permafrost Terrifies Scientists

Modeling Arctic Feedback Loops and Climate Tipping Points

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-ESS2-6, HS-LS2-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-6, HS-LS2-3

Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a quantitative computational model of the permafrost carbon feedback loop, testing scenarios to identify tipping point conditions.
Disciplinary Core Idea	**ESS2.D Weather and Climate / LS2.B Cycles of Matter and Energy Transfer** — Carbon cycles between the atmosphere, geosphere, and biosphere through processes including decomposition, absorption, and photosynthesis, with feedback loops that can amplify or dampen changes.
Crosscutting Concept	**Stability and Change** — Students model how a stable system (frozen permafrost storing carbon for millennia) can be pushed past a tipping point into a new, self-reinforcing state of change.

CAST Testing Alignment

MULTIPLE CHOICE:

Arctic temperatures have risen 3 degrees Celsius above pre-industrial levels. Based on your model, which feedback component would show the most dramatic change first?

CONSTRUCTED RESPONSE:

Using your model, explain why the permafrost carbon feedback is described as a 'carbon bomb' and why scientists fear it could make climate change irreversible even if all human emissions stopped. Reference at least three model components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Beneath the frozen ground of Siberia, Alaska, and northern Canada lies a carbon time bomb that dwarfs all human emissions combined. Permafrost — ground frozen continuously for at least two years, some of it for hundreds of thousands of years — stores approximately 1,500 billion tons of organic carbon. That's nearly twice the carbon in the entire atmosphere. As the Arctic warms at twice the global rate, this ancient carbon is beginning to thaw.

What Is Permafrost?

Permafrost is ground that remains at or below 0 degrees Celsius for at least two consecutive years. It covers approximately 25% of the Northern Hemisphere's land surface — an area larger than all of North and South America combined. Some permafrost in Siberia has been frozen for 700,000 years. During all that time, dead plants and animals accumulated in the frozen soil without decomposing, creating an enormous reservoir of organic carbon trapped in a natural freezer.

The Thaw Cascade

When permafrost thaws, several interconnected processes begin simultaneously. Soil microorganisms that have been dormant for millennia reactivate and begin decomposing the ancient organic material, producing CO2 in aerobic (oxygen-rich) conditions and methane in anaerobic (oxygen-poor) conditions. Methane is 80 times more potent than CO2 as a greenhouse gas over 20 years. The ground physically collapses as ice lenses melt, creating thermokarst lakes that accelerate further thawing. These lakes become concentrated sources of methane that bubbles visibly to the surface.

The Feedback Loop That Terrifies Scientists

The permafrost carbon feedback is a textbook positive feedback loop. Human emissions warm the atmosphere. The warmer atmosphere thaws permafrost. Thawed permafrost releases CO2 and methane. These additional greenhouse gases warm the atmosphere further. The warmer atmosphere thaws more permafrost. Each cycle amplifies the last. Current estimates suggest this feedback could add 0.3 degrees Celsius of additional warming by 2100 under high-emission scenarios — on top of the warming from human emissions. The critical question is whether this feedback has a tipping point beyond which it becomes self-sustaining.

Can Anything Stop It?

Two natural processes partially counteract the permafrost feedback. Oceans absorb about 25% of atmospheric CO2 through dissolution and biological uptake (phytoplankton). Vegetation expansion into newly thawed Arctic regions absorbs some carbon through photosynthesis — shrubs and trees are moving northward as conditions warm. However, studies consistently show that these sinks cannot match the rate of carbon release from thawing permafrost. The carbon was accumulated over tens of thousands of years but is being released over decades — an asymmetry that overwhelms the planet's absorption capacity.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the permafrost carbon system: Permafrost Temperature, Methane Release Rate, Microbial Decomposition, Atmospheric CO2, Ocean Absorption, Vegetation Growth, and Feedback Amplification — with Atmospheric CO2 as the external driver.

E **Establish Relationships**

Students determine that Atmospheric CO2 raises global temperature, increasing Permafrost Temperature. Thawing activates Microbial Decomposition and Methane Release Rate, which increase Atmospheric CO2 further. Ocean Absorption and Vegetation Growth provide partial offsetting, but Feedback Amplification tracks the net acceleration.

V **Visualize & Model**

Students build a computational model showing the positive feedback loop and the partial negative feedback from carbon sinks, tracking whether the net effect pushes toward a tipping point.

E **Evaluate Outcomes**

Students run current trajectory, aggressive emissions cuts, and methane burst scenarios to determine whether the tipping point can be avoided and what happens if it is crossed.

R **Revise & Extend**

Students add Thermokarst Formation, Subsea Permafrost, or Albedo Change to explore additional feedback mechanisms and their contribution to tipping point dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"There's a bomb buried under the Arctic. Not a weapon — something far more dangerous. 1,500 billion tons of carbon frozen in the ground, and the fuse is already lit."*

Watch For

Show a satellite time-lapse of Arctic ice retreat over 30 years. Then reveal the permafrost carbon number. Let the scale register.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling the feedback loop that keeps climate scientists awake at night — the one that could make climate change unstoppable."*

Watch For

Have students read objectives. Pre-teach 'positive feedback loop' and 'tipping point.' Quick-write: What's the difference between a feedback that stabilizes and one that amplifies?

Slide 3: Big Question

3 min

Say: *"Twice the carbon in the entire atmosphere is frozen under your feet if you're standing in Siberia. What happens when that freezer stops working?"*

Watch For

Visual comparison: Stack representing atmospheric carbon next to stack representing permafrost carbon. Which is bigger? Students react.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model a system with a terrifying property — it can reach a point where it keeps going even if we try to stop it."*

Watch For

Preview each LEVER step. Emphasize that this model contains both positive (amplifying) and negative (dampening) feedback loops.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components, two types of feedback, and one critical question: can we stop it?"*

Watch For

Guide component sorting. Discuss why Atmospheric CO2 is external (driven by human emissions policy) while the Arctic system responds internally.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the loop: CO2 warms the air, air thaws the ground, ground releases more CO2. Now find what slows it down — and ask if it's enough."*

Watch For

Help students map both the positive feedback (warming-thawing-release) and the negative feedback (ocean absorption, vegetation growth). Calculate the net effect.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three futures: business as usual, aggressive cuts, and a methane burst. Only one of them has a chance of avoiding the tipping point."*

Watch For

Students predict outcomes first. Run all three scenarios. Identify the tipping point threshold in each. Discuss which scenario is most realistic.

Slide 8: Discoveries

5 min

Say: *"Your model just showed why this is called a carbon bomb. Once the feedback becomes self-sustaining, it doesn't matter what humans do."*

Watch For

Lead discussion connecting model findings to current Arctic observations. Show real methane bubble data from Siberian lakes.

Slide 9: STEM Challenge

12 min

Say: *"Scientists need to know exactly when the tipping point is approaching. Design the warning system."*

Watch For

Groups design Arctic monitoring networks. Must specify sensors, locations, power sources, data analysis methods, and warning thresholds.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atmospheric CO2
- **INTERNAL:** Permafrost Temperature, Methane Release Rate, Microbial Decomposition, Ocean Absorption, Vegetation Growth, Feedback Amplification

Relationships

- **Atmospheric CO2 to Permafrost Temperature:** POSITIVE (+) — Higher Atmospheric CO2 strengthens the greenhouse effect, raising global and especially Arctic temperatures (Arctic amplification causes 2-3x faster warming at poles), which increases Permafrost Temperature toward the thawing threshold.
- **Permafrost Temperature to Microbial Decomposition:** POSITIVE (+) — As Permafrost Temperature rises above 0 degrees Celsius, previously frozen organic material becomes available to soil microorganisms. Higher temperatures accelerate microbial metabolic rates, increasing decomposition of ancient carbon stores.
- **Microbial Decomposition to Atmospheric CO2:** POSITIVE (+) — Active Microbial Decomposition converts ancient organic carbon into CO2 (in aerobic conditions) and methane (in anaerobic conditions), both of which enter the atmosphere and increase greenhouse gas concentrations — completing the positive feedback loop.

Key Discoveries

- The permafrost carbon feedback is a positive feedback loop — thawing releases greenhouse gases that cause more warming that causes more thawing, creating a self-amplifying cycle
- Ocean Absorption and Vegetation Growth provide negative feedback that slows the cycle, but the carbon being released from permafrost vastly exceeds what these sinks can absorb
- The system has a tipping point beyond which the feedback becomes self-sustaining — meaning permafrost carbon release would continue even if all human emissions stopped immediately
- The 1,500 billion tons of carbon in permafrost is a legacy of tens of thousands of years of organic accumulation — its release over decades rather than millennia would overwhelm Earth's carbon cycle

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Permafrost is just frozen dirt"	Permafrost is frozen ground that contains enormous amounts of organic carbon — dead plants and animals accumulated over tens of thousands of years. In some locations, permafrost is 30-50% organic material by weight. When it thaws, this material becomes available for microbial decomposition, releasing greenhouse gases. It is not just dirt — it is the largest terrestrial carbon reservoir on Earth.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Planting trees can offset permafrost emissions"	While Vegetation Growth does absorb some CO2, the rate of carbon release from thawing permafrost vastly exceeds what new Arctic vegetation can absorb. Trees grow slowly in Arctic conditions and cover only a fraction of the thawing area. Furthermore, dark trees absorb more solar radiation than white snow (albedo effect), potentially causing additional local warming. The math simply doesn't work — you cannot plant enough trees to offset 1,500 billion tons.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The Arctic has always been frozen so it will stay frozen"	The Arctic has not always been frozen. During the Eocene epoch (50 million years ago), the Arctic was ice-free with palm trees and crocodiles near the North Pole. Permafrost formed during ice ages and has persisted through warmer interglacial periods — but it has never experienced the rate of warming occurring today. Current warming is happening 10-100 times faster than natural cycles, giving permafrost no time to gradually adjust.